Financial Foundation Models with a Third Attention: Recovering Relational Signal at Fine-Tuning Time

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ABSTRACT

Financial foundation models (FFMs) pretrain a transformer on raw event sequences and adapt it to downstream tasks such as fraud detection with a lightweight head. Architecturally they rest on *two* attentions: one *within* an event, over its fields, and one *across* an account's own history. This makes them structurally blind to *relational* signal that lives in a shared entity's activity *across other accounts* — a merchant compromised right now, an IP driving many devices, a product going viral. We add a *third* attention: a cross-sequence encoder that, at *fine-tuning time*, lets a target event attend over the recent events of its shared entity from other sequences. Across four datasets — a controlled synthetic benchmark, two public fraud corpora (TalkingData, IEEE-CIS), and a non-fraud e-commerce conversion task (Retailrocket) — the third attention yields a consistent +0.04 to +0.07 PR-AUC gain over standard fine-tuning wherever a relational signal exists, and *collapses to a zero gain* on a synthetic control that contains no relational signal — isolating the improvement as cross-sequence information rather than added model capacity. On TalkingData it further beats a strong gradient-boosted-tree baseline. Because the module is added at adaptation and generalises beyond fraud, it offers a practical, composable route to relational FFMs.

KEYWORDS

financial foundation models, fraud detection, event-sequence transformers, relational learning, cross-attention, gradient-boosted trees

1 INTRODUCTION

Fraud detection and related risk tasks at scale are still dominated by gradient-boosted decision trees (GBDTs) over hand-engineered features [2, 6]. A compelling alternative, mirroring foundation models elsewhere [1], is to pretrain a transformer on raw event sequences with a self-supervised objective and adapt it to downstream tasks with a lightweight head — a *financial foundation model* (FFM) [9, 10, 14]. Each transaction is tokenised into field–value tokens; a set transformer encodes the fields of an event, and a temporal transformer encodes an account's history of such events; adaptation reads the final-position embedding with a linear probe or a light fine-tune.

In effect these models use *two attentions*: one *within* an event, over its fields, and one *across* an account's own history. Our observation is that this design sees only one entity in isolation and is therefore **structurally blind to relational signal** — patterns that live in a shared entity's activity *across* accounts. A compromised merchant produces a burst of fraud spread over many cards; a fraud-farm IP drives many devices; a product goes viral across many shoppers. None of these are visible to a model that attends only within a single account's sequence, no matter how it is trained.

We add a *third*, **cross-sequence attention**: for a target event at a shared entity (merchant, IP, item, ...), the model attends over that entity's recent events from *other* sequences, encoded by the same event encoder. Crucially we add it at *fine-tuning time*, on top of a pretrained two-attention backbone, rather than changing the pretraining recipe. This keeps the FFM intact and makes the module a drop-in adaptation choice.

Our contributions are:

- **A third attention for FFMs (§3).** A cross-sequence encoder that lets a target event attend over a shared entity's recent cross-account events, added at fine-tuning time and trained end-to-end with a linear head.
- **Consistent relational gains (§5).** Across three datasets where a relational signal exists, the third attention improves PR-AUC by +0.04 to +0.07 over standard fine-tuning of the same backbone.
- **An isolation control.** On a synthetic benchmark with *no* relational signal, the gain *collapses to zero* — the same added capacity yields nothing — establishing that the improvement is the cross-sequence signal, not extra parameters.
- **Beating a strong tabular baseline, and generalisation beyond fraud.** On TalkingData the fine-tuned FFM with the third attention exceeds a strong GBDT; and on a non-fraud e-commerce conversion task (Retailrocket) the same module delivers the same relational lift, showing the idea is not fraud-specific.

All code, the synthetic generator, dataset adapters, and per-experiment result files are released for reproducibility.

2 RELATED WORK

Event-sequence and tabular transformers. Encoder-only, masked-pretrained transformers [3, 13] over tokenised tabular/event sequences are well established: BEHRT for medical records [7] and TabBERT/TabFormer for transactions [10] introduced the hierarchical field-then-sequence design our backbone follows, and related tabular transformers add contextual and row-attention encodings [4, 5, 11]. The neural architecture we build on is therefore not itself novel; our object of study is what the *two-attention* recipe misses relationally, and how to add it cheaply.

Industrial financial foundation models. Two recent industrial systems are the closest prior work. PRAGMA [9] pretrains a masked transformer on heterogeneous banking event sequences and serves credit scoring, fraud, and lifetime-value prediction from one representation read by a linear probe or a light fine-tune — the recipe we reproduce at laptop scale. TREASURE [14] encodes payment-network transactions with dedicated static/dynamic modules and ingests payment-network signals alongside consumer behaviour. Both encode a *single entity's own event stream*; neither reads, at inference, the concurrent activity of a shared entity across *other* accounts. Our cross-sequence attention is orthogonal and composable — it augments such a backbone rather than replacing it.

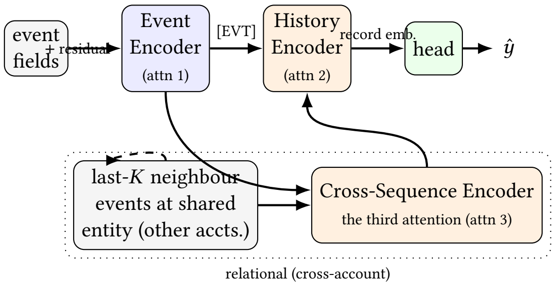

Figure 1: A standard FFM has *two* attentions: over an event's fields (Event Encoder) and over an account's own history (History Encoder). We add a *third*: a Cross-Sequence Encoder that attends over a shared entity's recent events from *other* accounts, encoded by the same Event Encoder, and adds the result as a residual to the target's record embedding.

Adaptation: probing vs. fine-tuning. Adapting a pretrained encoder ranges from a frozen linear probe over its embeddings to full fine-tuning. A frozen probe measures what the self-supervised objective already preserved; fine-tuning lets the task reshape the backbone. We use both as baselines and show that the relational signal is recovered only when the third attention is present, whether frozen features are read or the backbone is fine-tuned.

GBDTs for fraud. Gradient-boosted trees [2, 6] with client- and entity-level aggregate features remain the production and competition standard for tabular fraud, and are a demanding baseline; we compare against a strong GBDT given per-entity causal aggregates on the identical evaluation subsample.

3 METHOD

3.1 The two-attention FFM backbone

We reproduce a standard FFM at laptop scale (≈10M parameters).

Tokenisation. Each event (a transaction, click, or page view) is a set of (field, value) pairs. Categorical values are mapped to per-field vocabularies; numeric values (e.g. amount) are discretised into learned buckets; two calendar fields (hour of day, day of week) are derived from the event timestamp. A learned `[EVT]` token is prepended to each event's token set.

Two attentions. An **event encoder** — a set transformer with no positional encoding — contextualises an event's field tokens and returns its `[EVT]` vector, a per-event embedding. A **history encoder** — a bidirectional transformer with rotary positional encoding [12] on *continuous event time* — runs over an account's sequence of `[EVT]` vectors and returns a record embedding h_t per event. These are the two attentions: within an event, and across an account's own history.

Pretraining. We pretrain with masked modelling: a fraction of (field, value) tokens are masked and reconstructed by per-field heads, a self-supervised objective requiring no labels. Concretely the backbone has d_{model}=256, 8 attention heads, feed-forward width 1024, and a context window of 128 events. We train each dataset's backbone for 30000 steps with AdamW [8], a peak learning rate of 3e-4, 500 warm-up steps and cosine decay, batch size 128, in `bf16` on a single NVIDIA L40S GPU. Figure 2 shows the masked-modelling loss plateauing on every dataset, confirming the backbone is converged before any adaptation.

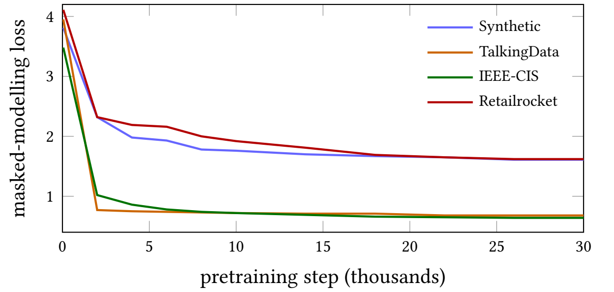

Figure 2: Masked-modelling pretraining loss for each dataset's backbone. Every curve plateaus well before 30000 steps, so each FFM is converged before any adaptation.

3.2 The third attention: a cross-sequence encoder

A per-account model never sees a shared entity's cross-account state. We add a third attention that supplies exactly this. For a target event on account a at shared entity e (merchant, IP, item) and time t, with record embedding q=$h_t \in \mathbb{R}^d$ from the history encoder, we gather the neighbour set $\mathcal{N}_{e,t}$ = {the last K events at e with time strictly $< t$} — causal, across *all* accounts, $|\mathcal{N}_{e,t}| \le K$. Each neighbour event j is encoded by the *shared* event encoder E into its `[EVT]` token, to which we add a learned recency embedding of its lag:

$$\mathbf{z}_j \;=\; E(x_j) + \rho\big(\log(t - t_j)\big). \tag{1}$$

An optional shallow transformer T mixes the K neighbours, $\{\mathbf{n}_j\} = T(\{\mathbf{z}_j\})$. The target *queries* them by multi-head cross-attention with a padding mask m_j (0 for a real neighbour, $-\infty$ for padding):

$$Q = W_Q\, q, \quad K_j = W_K\, \mathbf{n}_j, \quad V_j = W_V\, \mathbf{n}_j, \tag{2}$$

$$\alpha_j = \operatorname{softmax}_j\Big(\frac{Q^\top K_j}{\sqrt{d_h}} + m_j\Big), \qquad u = \textstyle\sum_j \alpha_j\, V_j. \tag{3}$$

The attention output is used only when the target has at least one in-window neighbour, so an empty neighbourhood contributes an exact zero rather than a masked-softmax NaN:

$$p \;=\; \mathbb{1}\big[\mathcal{N}_{e,t} \neq \varnothing\big] \cdot \operatorname{LayerNorm}\big(W_O\, u\big). \tag{4}$$

Alongside the attention path, the encoder carries an always-on readout of a precomputed, causal windowed velocity $c \in \mathbb{R}^{d_c}$ of the shared entity (the log count of prior same-entity events within fixed time windows), projected and added:

$$\operatorname{xseq}(q, \{\mathbf{n}_j\}, c) \;=\; p \;+\; \operatorname{LN}\big(\operatorname{GELU}(W_c\, c)\big), \qquad h'_t = h_t + \operatorname{xseq}\,. \tag{5}$$

The result is a residual added to the target's record embedding, which the head then reads (Algorithm 1). The two internal paths — attention over raw neighbours and the velocity readout — are complementary; we treat the module as a single unit and leave a formal decomposition of their contributions to future work. Neighbours are the most recent K prior events at e (temporal recency,

Algorithm 1 Cross-sequence attention for one target event

Require: target embed. $q{=}h_t$, entity e, time t, velocity c, window K; shared encoder E

$\mathcal{N} \leftarrow$ last K events at e with time $< t$ ▷ causal, all accounts
if $\mathcal{N} = \varnothing$ **then**
  $p \leftarrow \mathbf{0}$
else
  for $j \in \mathcal{N}$ **do**
    $\mathbf{z}_j \leftarrow E(x_j) + \rho(\log(t - t_j))$
  end for
  $p \leftarrow \mathrm{LN}\big(W_O\ \mathrm{MHA}(q, T(\{\mathbf{z}_j\}))\big)$
end if
return $h_t + p + \mathrm{LN}(\mathrm{GELU}(W_c\, c))$

not similarity selection); their embeddings are recomputed each step so gradients flow into the shared encoder. Throughout we use $K{=}16$, one mixing layer, and $d_c{=}2$ velocity windows.

3.3 Adaptation strategies

We evaluate three ways to adapt the pretrained backbone to a binary target, plus the tabular baseline below. All read the target event's final record embedding with a single linear head ($\mathbb{R}^d \to \mathbb{R}$) and train a class-weighted binary cross-entropy over as-of-date windows (a window ending at each target event, no future leakage), with AdamW at learning rate 3e-4, `bf16`, a single seed.

- **Linear probe.** The backbone is *frozen*; only the linear head is trained. This measures what masked pretraining already preserved.
- **Fine-tune.** The backbone is unfrozen and trained end-to-end with the head. This lets the task reshape the two-attention representation.
- **Fine-tune + cross-attention.** The third attention (§3.2) is added and the whole model — backbone, cross-sequence encoder, and head — is fine-tuned end-to-end from the same checkpoint. The cross-sequence encoder's weights are randomly initialised (it is not part of pretraining) and learned during adaptation.

We fine-tune for 8 epochs on the synthetic data and 6 on the larger real corpora, with batch size 128 (synthetic) or 256 (real).

3.4 Gradient-boosted-tree baseline

As a strong tabular reference we train LightGBM [6] (400 trees, 64 leaves, learning rate 0.05, subsample and column-sample 0.8, class-balanced) on the same stratified evaluation subsample as the FFM, so PR-AUC is apples-to-apples. Its features are the per-event fields the event encoder sees *plus* the standard causal aggregates a practitioner engineers for both the sequence entity and the shared entity: prior count, windowed velocity, mean amount, and recency (seconds since that entity's previous event). It is not given raw entity ids, which would leak a label-derived per-entity rate. Given extreme class imbalance we report PR-AUC (average precision) throughout.

Table 1: Datasets. Response rate is the positive fraction on the stratified evaluation subsample.

Dataset	Seq. entity	Shared entity	Resp. rate
Synthetic (control)	card	merchant	0.024
TalkingData	user	IP	0.017
IEEE-CIS	card	region (addr)	0.042
Retailrocket	visitor	item	0.112

4 DATASETS

We evaluate on four datasets (Table 1): one controlled synthetic benchmark that acts as an isolation control, two public fraud corpora, and one non-fraud e-commerce task. For each we define a *sequence entity* (whose events form a sequence the history encoder models) and a *shared entity* (the cross-account hub the third attention attends over). Because positives are rare, we evaluate on a stratified subsample — all positive events plus a capped random sample of negatives (seed 0) — and report the resulting base rate; PR-AUC is compared *within* a dataset across arms, not across datasets. All splits are causal (as-of-date), with no future leakage.

Synthetic control (no relational signal). We generate card transactions in which a card is compromised at a random time and fires a short *burst* of fraud on *its own* timeline at random merchants. The signal is thus entirely *per-account* (a card's own recent behaviour) and there is, by construction, *no* cross-account relational signal at the merchant. This is our control: a correctly specified relational module should add nothing here. We split by card (unseen cards: 80/10/10); the evaluation subsample has base rate 0.024.

TalkingData (click fraud). A large public mobile click log. The label `is_attributed` marks a click that led to an app install — the genuine minority among a flood of fraudulent clicks. The sequence entity is the user = (IP, device, OS); the shared cross-user entity is the **IP** (a fraud farm drives many users from one IP). A per-user model cannot see how active an IP is across *other* users. We use a 6M-event time-contiguous subset, split temporally (last 20% test), base rate 0.017.

IEEE-CIS (card fraud). A public card-not-present fraud corpus (590K transactions, label `isFraud`). The sequence entity is the card; the shared entity is the billing region (`addr`), shared across cards. Much of the fraud is a function of the *current* transaction's attributes rather than of account history, making it a demanding, largely per-transaction setting; split temporally, base rate 0.042.

Retailrocket (e-commerce conversion — non-fraud). A public e-commerce clickstream (2.76M events, 1.41M visitors). We predict *conversion*: the positive is an event of type add-to-cart or purchase, everything else negative (base rate 0.033 overall, 0.112 on the stratified subsample); the event *type* is held out as the label and never an input feature. The sequence entity is the visitor; the shared entity is the **item**, which trends across many visitors concurrently — a "going viral now" signal a per-visitor model cannot see. Sequences are short (median length 1, mean 2), so the item-level relational signal is the main lever. Split temporally. This is a non-fraud task included to test generalisation.

Table 2: Main result: PR-AUC on the stratified evaluation subsample (single seed). Columns are the GBDT baseline and the three adaptation strategies of the *same* pretrained backbone. Δ is the gain of the third attention over standard fine-tuning. Bold marks the best adaptation arm; * marks where it also beats the GBDT.

Dataset	GBDT	probe	fine-tune	fine-tune + cross-attn	Δ
Synthetic (control)	0.849	0.793	**0.829**	0.823	−0.006
TalkingData	0.405	0.486	0.678	**0.714***	+0.036
IEEE-CIS	0.294	0.163	0.167	**0.230**	+0.063
Retailrocket	0.630	0.344	0.467	**0.534**	+0.067

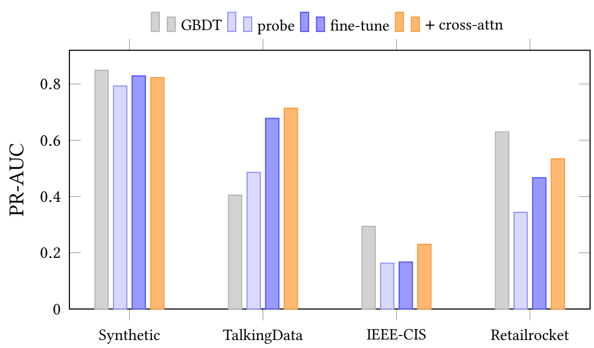

Figure 3: Main results (PR-AUC, single seed). Within each dataset, probe → fine-tune → +cross-attention is monotonic on the three relational datasets and flat on the synthetic control (no relational signal). The third attention beats the GBDT on TalkingData; on IEEE-CIS and Retailrocket it lifts the FFM but the GBDT remains ahead. Base rates differ across datasets, so bars are comparable within a dataset, not across.

5 RESULTS

Table 2 and Figure 3 report the full result. Three findings.

The third attention gives a consistent relational gain. On all three datasets that contain a cross-account signal, adding cross-sequence attention improves PR-AUC over standard fine-tuning of the same backbone — +0.036 on TalkingData, +0.063 on IEEE-CIS, and +0.067 on Retailrocket. The ordering probe → fine-tune → fine-tune+cross-attention is monotonic in every case: unfreezing the backbone helps, and the third attention helps further.

The gain is the relational signal, not added capacity. On the synthetic control — where the fraud is per-account by construction and no relational signal exists at the shared entity — the *same* added module yields no gain (-0.006; fine-tune 0.829 vs. cross-attention 0.823). Identical extra parameters that lift the three relational datasets do nothing here. This isolates the improvement as cross-sequence information rather than model capacity, and doubles as a specificity check: the module does not spuriously inflate a task it should not help.

It can beat a strong GBDT, and generalises beyond fraud. On TalkingData the fine-tuned FFM with the third attention reaches 0.714, clearing the strong GBDT baseline (0.405): here the FFM models a user's own click sequence, a signal the tree lacks, and the third attention adds the cross-user IP signal on top. On IEEE-CIS and Retailrocket the FFM trails the GBDT (0.230 vs. 0.294; 0.534 vs. 0.630) — there the relational signal is close to a scalar velocity that a tree exploits efficiently (§6) — yet the third attention still delivers its relational lift. Notably Retailrocket is a *non-fraud* conversion task: the same module, unchanged, recovers the "item trending now" signal, showing the approach is not fraud-specific.

6 DISCUSSION

Two attentions are not enough. A standard FFM attends within an event and across an account's own history, and is therefore blind to a shared entity's cross-account state. Our results show this is a real and recoverable gap: across three datasets with a relational signal, a cross-sequence attention added *at fine-tuning time* improves PR-AUC by +0.04 to +0.07 over fine-tuning the identical backbone, while adding the same module to a control with no relational signal changes nothing. Adapting the backbone with a third attention is thus an efficient, low-friction way to make an existing FFM relational — no change to the pretraining recipe, just a different adaptation choice.

When the FFM beats the tree, and when it does not. The FFM with the third attention beats a strong GBDT on TalkingData but trails it on IEEE-CIS and Retailrocket. The pattern is consistent: a GBDT is near-optimal at thresholding a *scalar* feature, so when the relational signal is essentially a velocity magnitude (IEEE-CIS's entity aggregates, Retailrocket's item popularity) the tree exploits it more efficiently than a neural residual readout on rare, imbalanced data. The FFM wins where it contributes something a tree cannot engineer from aggregates — the *sequential* structure of an account's own events, as on TalkingData, where the third attention then adds the cross-account signal on top. The takeaway is not "FFM beats GBDT" but that the third attention *recovers relational signal the two-attention FFM cannot see*, and that whether that clears a tree depends on how sequential the task is.

What relational signal the module still misses. That the FFM trails the tree on two datasets points to signal our cross-sequence encoder does not yet capture. It attends over the last K recent events at *one* shared entity, so it sees a recency-windowed neighbourhood and its velocity, but not: (i) *selection* under camouflage — when a fraudulent entity's neighbours are mostly legitimate, raw attention is diluted and similarity-based neighbour selection is needed; (ii) *multi-hop* or multi-entity relations beyond a single hub; and (iii) longer-range structure outside the K-window. These are natural extensions of the same module.

Productisation: one backbone, task-specific cross-attention. Because the third attention is added at adaptation, a single frozen (or lightly fine-tuned) backbone can serve many tasks, each with its *own* cross-attention head keyed on the entity that matters for that task — merchant or IP for fraud, item for recommendation, region for credit. The module is not one-size-fits-all: the choice of shared entity, window size, and the relative role of the attention and velocity paths should be tuned per downstream task. This flexibility — a composable relational adapter over a shared FFM — is, we argue, the practical form in which relational signal should enter a deployed financial foundation model.

Limitations. We report a single seed at laptop scale (≈10M parameters); the third attention is trained only at fine-tuning time, not pretrained; and our matched GBDT lacks hand-built per-sequence aggregates on some datasets, so the FFM's margin partly reflects sequence modelling. Pretraining the third attention with a relational self-supervised objective, and a formal decomposition of its attention and velocity paths, are the clearest next steps.

7 CONCLUSION

Financial foundation models rest on two attentions — within an event and across an account's own history — and are structurally blind to relational signal that lives in a shared entity's activity across other accounts. We showed that a *third* attention, a cross-sequence encoder added at fine-tuning time, recovers this signal: it yields a consistent +0.04 to +0.07 PR-AUC gain over standard fine-tuning across two fraud datasets and a non-fraud e-commerce task, collapses to no gain on a synthetic control with no relational signal, and on one dataset beats a strong gradient-boosted-tree baseline. Because it is a composable adapter over an unchanged backbone and generalises beyond fraud, the third attention is a practical route to relational financial foundation models. Pretraining it with a relational objective, and extending it with neighbour selection and multi-hop structure, are promising next steps.

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